

Activity 3: AI-Assisted Build Failure Analysis and Troubleshooting

Purpose

Learn how to use AI to analyze build and runtime errors, identify root causes, interpret error messages, and recommend appropriate solutions before implementing fixes.

Pre-requisites

- Completion of Activities 1 and 2
- Working SmartBank EMI Calculator application
- VS Code with Cline configured

Scenario

During development of the SmartBank EMI Calculator, several build and runtime errors have been reported by different developers. Before implementing any fixes, the development team wants to understand the root cause of each issue.

As a developer, your responsibility is to use AI to analyze error messages, explain their causes, suggest possible solutions, and recommend preventive measures.

Note: The objective of this activity is to learn how AI assists in troubleshooting build and runtime issues. You are not required to execute the application or reproduce the errors.

Tasks

Task 1: Analyze the Error Message

Review an error message and understand what it means.

Suggested Prompt

Act as a Senior Software Engineer.

Analyze the following build or runtime error.

Explain:

- What the error means
- Possible root cause
- Why the error occurred
- Which part of the application is likely affected

Explain the issue in simple language suitable for a beginner developer.

Do not suggest code changes yet.

<Paste the error message here>

Task 2: Identify the Root Cause

Use AI to investigate the most likely reason for the failure.

Suggested Prompt

Based on the following error message and the current Oracle JET application, identify the most likely root cause.

Explain:

- Why the error occurred
- Which file or component should be investigated
- Common developer mistakes that can lead to this error

Do not generate code.

<Paste the error message here>

Task 3: Recommend a Solution

Ask AI to suggest a structured approach to resolve the issue.

Suggested Prompt

Recommend a solution for the following error.

Provide:

- Step-by-step troubleshooting approach
- Possible fixes
- Best practices to avoid similar issues
- Things to verify after applying the fix

Do not modify the application code.

<Paste the error message here>

Task 4: Review the AI Recommendation

Evaluate whether the AI's suggestions are practical and complete.

Suggested Prompt

Review the proposed troubleshooting recommendations.

Evaluate:

- Accuracy
- Completeness
- Practicality
- Any missing steps
- Alternative approaches

Summarize whether the recommendations are suitable for resolving the issue.

Sample Errors for Practice

Use any one of the following errors during the activity.

Error 1 – JavaScript Syntax Error

```
Uncaught SyntaxError: Unexpected token '}'
```

Error 2 – Missing Oracle JET Module

```
Error: Cannot find module 'ojs/ojbutton'
```

Error 3 – API Connection Error

```
TypeError: Failed to fetch
```

Error 4 – JSON Parsing Error

```
Unexpected token < in JSON at position 0
```

Error 5 – Undefined Variable

```
ReferenceError: loanAmount is not defined
```

Error 6 – Null Object Reference

```
TypeError: Cannot read properties of null
```

Expected Deliverables

By the end of this activity, you should have:

- Error analysis report
- Root cause analysis
- AI-generated troubleshooting recommendations
- Best practices to prevent similar issues